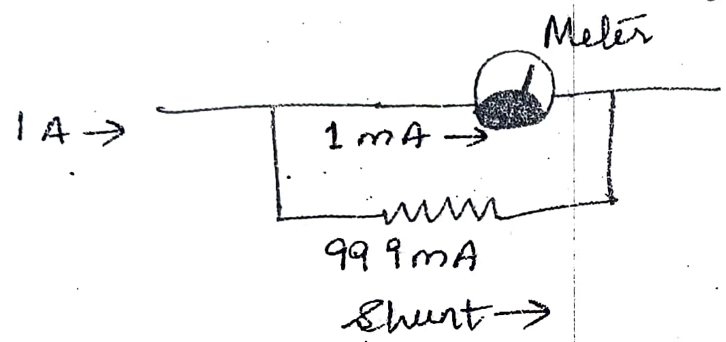
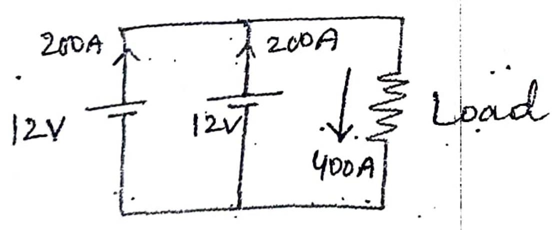
## Applications:

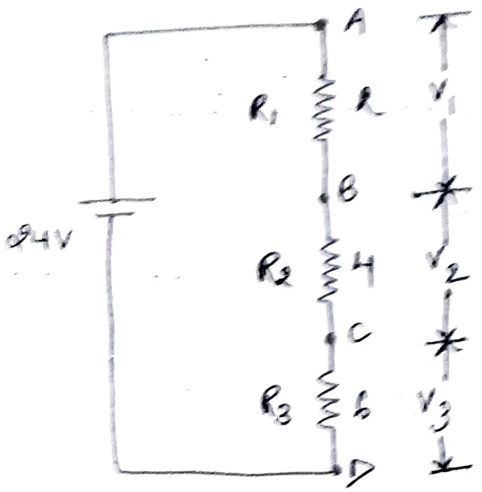
(i) A low resistor, called a shunt is connected in parallel with an ammeter to increase the current range of the meter. If shunt is not used, the ammeter is able to measure current upto 1mA. Use of shunt permits to measure current upto 1A. Shunt increases the range of ammeter.




(ii) Identical voltage sources may be connected in parallel to provide a greater current capacity.



# Voltage Divider Rule :




Total Resistance

$R = R_1 + R_2 + R_3$

Acc. to Voltage Divider Rule

$V_1 = \frac{R_1}{R} \times V$

$V_2 = \frac{R_2}{R} \times V$

$V_3 = \frac{R_3}{R} \times V$

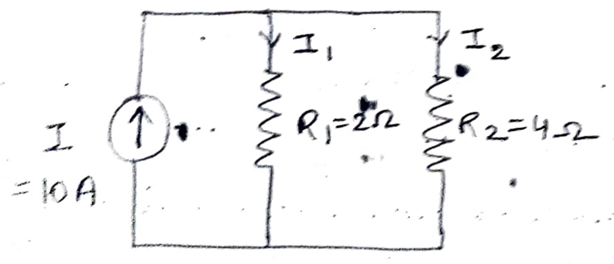
eg. $R = 12\,\Omega$

$V_1 = \frac{2}{12} \times 24 = 4V$

$V_2 = \frac{4}{12} \times 24 = 8V$

$V_3 = \frac{6}{12} \times 24 = 12V$

# Current Divider Rule :




$I_1 = \frac{R_2 \times I}{R}$

$I_2 = \frac{R_1 \times I}{R}$

$\boxed{R = R_1 + R_2}$

$R = 6\,\Omega$

$I_1 = \frac{4}{6} \times 10 = \frac{20}{3} A$

$I_2 = \frac{2}{6} \times 10 = \frac{10}{3} A$

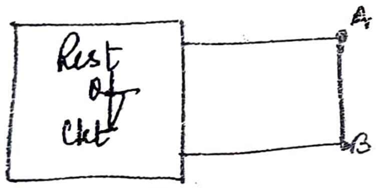
# SHORT & OPEN CIRCUITS

**SHORT CIRCUIT:** when two points of circuit are connected together by a thick metallic wire.

**SHORT:** zero resistance

(i) no voltage can exist across it because $V = IR = I \times 0$, $V = 0$

(ii) current through it (short circuit current) is very large ($\infty$)



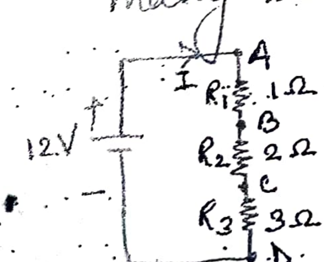
$V_{AB} = 0$

$I_{sc} = I_{AB}$

$I_{sc} = \infty$

## SHORTS IN A SERIES CKT

A dead (or solid) short has almost zero resistance, it causes the problem of excessive current which in turn, causes power dissipation to increase many times and circuit components to burn out.

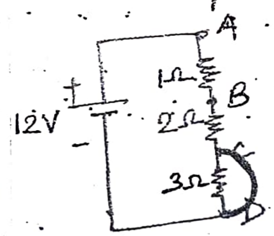


$V = 12V$

$R = 6\Omega$

$I = 2A$

$P = I^2R = 24W$

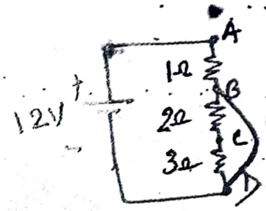


$V = 12V$

$R = 2+1 = 3\Omega$

$I = 4A$

$P = 48W$



$V = 12$

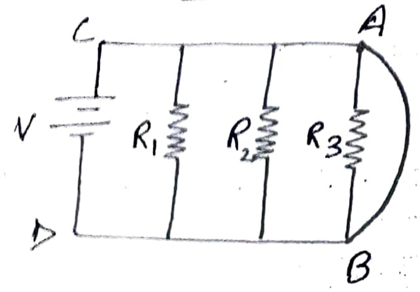
$R = 1\Omega$

$I = 12A$

$P = 144W$

→ Due to short circuit, current become 6 times the normal value.

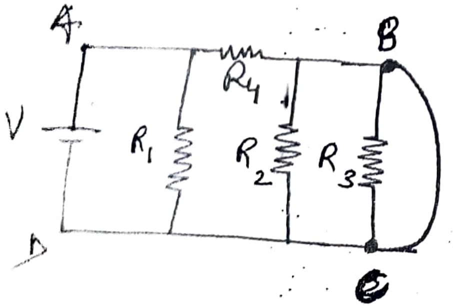
## SHORTS IN PARALLEL CIRCUIT




→ Not only $R_3$ is short circuited but both $R_1$ and $R_2$ are also shorted out i.e short across one branch means short across all branches.

→ There is no current in shorted resistors. If there were three bulbs they will not glow.

→ The shorted components are not damaged. For eg. if we had three bulbs, they would glow again when ckt is restored to normal conditions by removing the short circuited.




→ In this fig., short-circuit across $R_3$ may short out $R_2$ but not $R_1$, since it is protected by $R_4$.

## OPEN CIRCUIT :

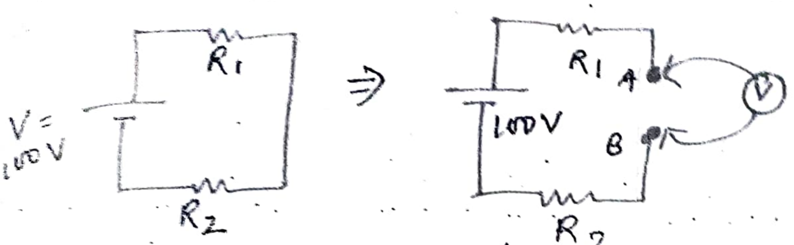
→ Two points are said to be open circuited when there is no direct connection between them.

→ Open : represents break in the continuity of the circuit.

Due to break : (I) Resistance b/w two points is infinite.

(II) No flow of current b/w two points.

# OPEN IN A SERIES CIRCUIT :-



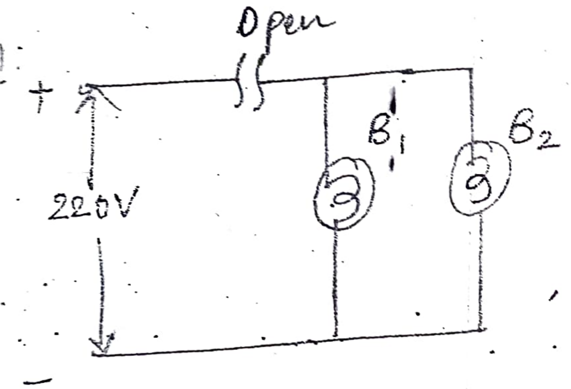

Effects produced due to open ckt :

① → Infinite resistance circuit current = 0
No voltage drop across $R_1$ & $R_2$.

② → Whole of the applied voltage (i.e. 100V) is felt across the 'open' i.e. across terminals A and B.

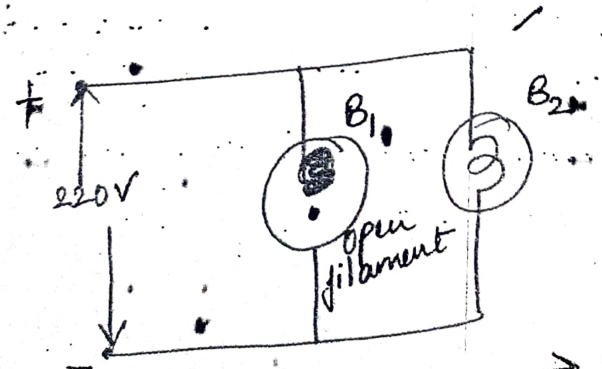
# OPEN IN PARALLEL CIRCUIT

case 1 :




→ An open in main line prevents flow of current to all branches.
Hence, neither of two bulb glows.
However, 220V (full applied voltage) is available across the open.

case 2 :




→ open has occurred in branch circuits of $B_1$.
No current in this branch, $B_1$ will not glow.

→ other bulb remains connected across supply, it would keep operating normally.

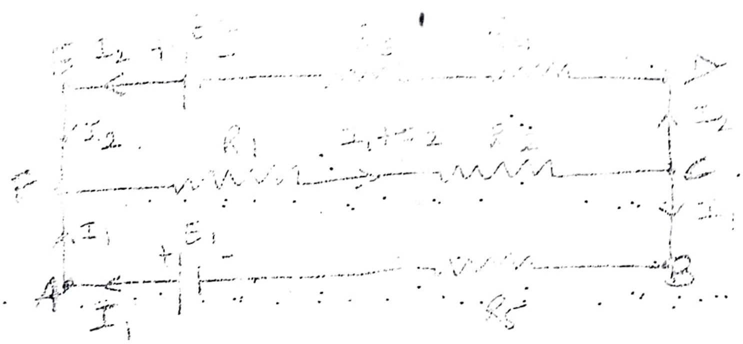
(2) KIRCHHOFF'S SECOND LAW or VOLTAGE LAW or MESH LAW

"The algebraic sum of the products of currents and resistances in each of the conductors in any closed path in a network plus the algebraic sum of the emfs in that path is zero."

$$\boxed{\sum I.R + \sum e.m.f = 0}$$

or

Acc to this law, "In any closed circuit or mesh the algebraic sum of emfs acting in that circuit is equal to the algebraic sum of the products of the currents and resistances of each part of the circuit. i.e. $\sum e.m.f = \sum IR$



In mesh AFCBA,

Apply KVL:

$$\underline{-E_1} + \underline{R_1(I_1+I_2) + R_2(I_1+I_2) + R_5 I_1} = 0 \quad (1^{st}\ deg^n)$$

$$E_1 = I_1 R_5 + R_1(I_1+I_2) + R_2(I_1+I_2) \quad (2^{nd}\ deg^n)$$

$$E_1 = I_1 R_5 + (R_1+R_2)(I_1+I_2)$$

In mesh FEDCF,

$$R_4 I_2 + R_3 I_2 - E_2 + R_1(I_1+I_2) + R_2(I_1+I_2) = 0$$

$$\underline{-E_2} + \underline{(R_4+R_3)I_2 + (R_1+R_2)(I_1+I_2)} = 0 \quad (1^{st}\ deg^n)$$

$$E_2 = (R_4+R_3)I_2 + (R_1+R_2)(I_1+I_2) \quad (2^{nd}\ deg^n)$$

# KIRCHHOFF'S LAWS

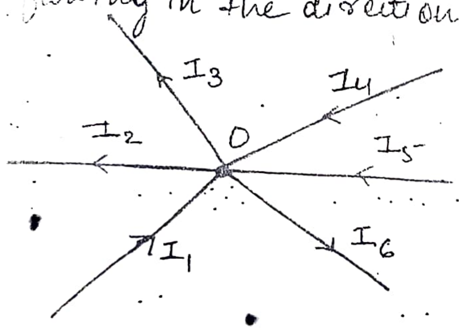
These laws describe the relationships among circuit voltages and circuit currents.
These laws are very helpful in determining the
→ equivalent resistance of a complex network, &
→ the current flowing in the various branches of the network.

Gustov Robert Kirchhoff derived two basic laws :

## (1.) KIRCHHOFF'S FIRST LAW or CURRENT LAW (KCL) or Point LAW

"According to this law in any network of wires carrying currents, the algebraic sum of all currents meeting at a point (or junction) is zero."

If $I_1, I_2, I_3, I_4, I_5$ & $I_6$ are the current meeting at junction O, flowing in the directions of arrowheads marked on them




$$\boxed{I_1 - I_2 - I_3 + I_4 + I_5 - I_6 = 0}$$

taking
incoming currents as +ve
& outgoing currents as −ve

or

"The sum of incoming currents towards any point is equal to the sum of outgoing currents away from that point."

$$\boxed{I_1 + I_5 + I_4 = I_3 + I_2 + I_6}$$

Incoming current = Outgoing current

In ② AFEDCBA

$$R_5 I_1 - E_1 + E_2 - R_3 I_2 - R_4 I_2 = 0$$

$$\underline{-E_1 + E_2} + \underline{R_5 I_1 - R_3 I_2 - R_4 I_2} = 0 \quad \text{(1st eqn)}$$

$$E_1 - E_2 = R_5 I_1 - (R_3 + R_4) I_2 \quad \text{(2nd eqn)}$$

## Sign Conventions

**1st method:** As discussed above.

→ The resistive drops in a mesh due to current flowing in clockwise direction must be taken positive drops.

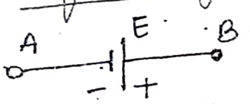
→ The resistive drops in a mesh due to current flowing in counter-clockwise direction must be taken as -ve drops.

→ Battery emf causing current to flow in clockwise direction in a mesh must be taken as positive emf.

→ Battery emf causing current to flow in counter-clockwise direction in a mesh must be taken as negative emf.

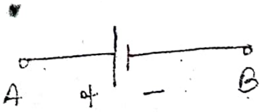
**2nd Method:**

→ Sign of Battery E.M.F.




(Rise in voltage
+E)

A rise in voltage should be given a +ve sign
i.e. if we go from -ve terminal of a battery to +ve terminal, there is a rise in potential.

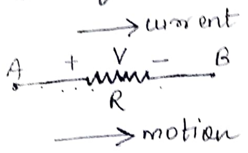



Fall in voltage
-E

A fall in voltage should be given a -ve sign.
i.e. if we go from +ve terminal to -ve terminal, there is a fall in voltage.

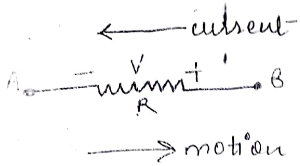
★ It is important to note that the sign of the battery e.m.f. is independent of the direction of current through that branch.

→ **Sign of IR drop:**




(Fall in voltage
$-V = -IR$)

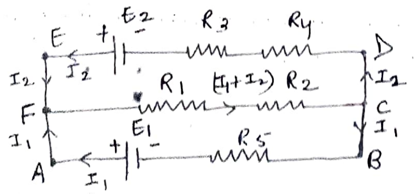
If we go through a resistor in the same direction as the current, then there is a fall in potential because current flows from higher to a lower potential.
Hence, this voltage fall should be taken as −ve.




(Rise in voltage
$+V = +IR$)

If we go in a direction opposite to that of the current.
So current flow: ← lower to higher
we proceed →
So there is rise in voltage
& it should be taken as +ve sig.

eg. consider the same ckt as discussed in previous example.




In mesh AFCBA

$E_1 - R_1(I_1+I_2) - R_2(I_1+I_2) - I_1$

$E_1 = I_1 R_5 + (R_1+R_2)(I_1+I_2)$

In mesh AFEDCBA

$-E_2 + (R_3+R_4)I_2 - R_5 I_1 + E_1 = 0$

$E_1 - E_2 = R_5 I_1 - (R_3+R_4)I_2$

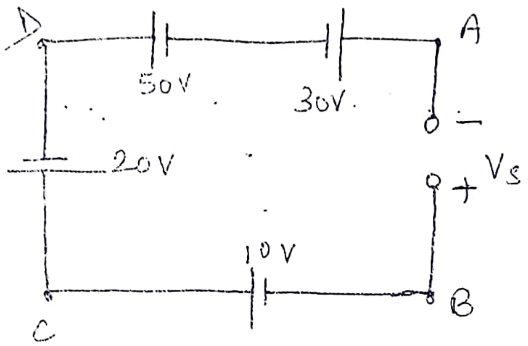
In mesh FEDCF

$E_2 - R_1(I_1+I_2) - R_2(I_1+I_2) - (R_3+R_4) = 0$

$E_2 = (R_1+R_2)(I_1+I_2) + (R_3+R_4)I_2$

# Questions Based On Kirchhoff's Law.

Q.1, What is the voltage $V_s$ across the open switch in the given ckt.




Apply KVL.

$$+V_s + 30 - 50 - 20 + 10 = 0$$

$$V_s + 40 - 70 = 0$$

$$V_s - 30 = 0$$

$\boxed{V_s = 30V}$ (According to 1st method)

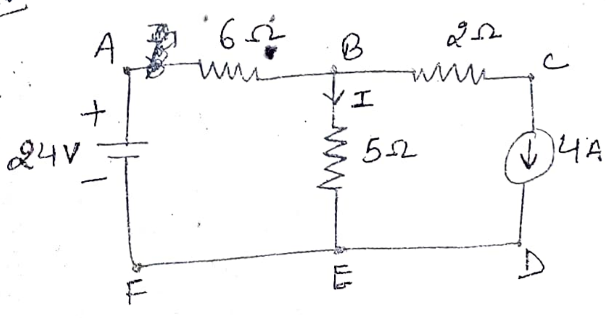
2nd method (sign convention)

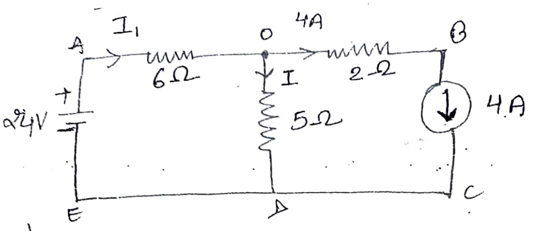
$$-V_s - 30 + 50 + 20 - 10 = 0$$

$$V_s = 20 + 10$$

$\boxed{V_s = 30V.}$

Q2: Find the current I in the ckt.






1st method  At node O  By using KCL

$$I_1 = I + 4A$$

$$I = I_1 - 4$$

$$= \frac{24}{6} - 4$$

$$= 4 - 4$$

$$\boxed{I = 0}$$

2nd method

$$I_1 = I + 4 \quad \text{(Acc. to KCL)}$$

Apply KVL, in mesh AODEA (For applying KVL first we have to find current in each branch of given mesh)

$$-24 + 6I_1 + 5I = 0$$

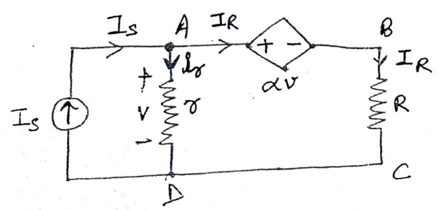
$$-24 + 6(I+4) + 5I = 0 \quad \text{(Put the value } I_1 = I + 4\text{)}$$

$$\cancel{-24} + 6I + \cancel{24} + 5I = 0$$

$$11I = 0$$

$$\boxed{I = 0}$$

Q-3. Obtain $I_R$ in terms of $I_s$.



Apply KCL at node A

$$I_s = i_r + I_R$$

Apply KVL in mesh ABCDA

$$+\alpha V + R I_R - V = 0.$$

we know $V = r\, i_r$

$$\alpha . r . i_r + R I_R - r i_r = 0$$

$$R I_R = r i_r - \alpha r i_r$$

$$= i_r (r - \alpha r)$$

$$R I_R = (I_s - I_R)(r - \alpha r)$$

$$R I_R = r I_s - \alpha r I_s - r I_R + \alpha r I_R$$

$$R I_R = (r - \alpha r) I_s + I_R (\alpha r - r)$$

$$I_R (R - \alpha r + r) = r(1-\alpha) I_s$$

$$I_R = \frac{r(1-\alpha) I_s}{R - r(\alpha - 1)}$$

$$I_R = \frac{r(1-\alpha) I_s}{R + r(1-\alpha)}$$

Q. 4

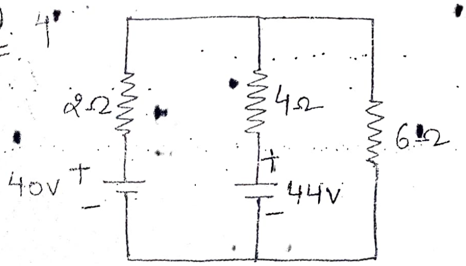
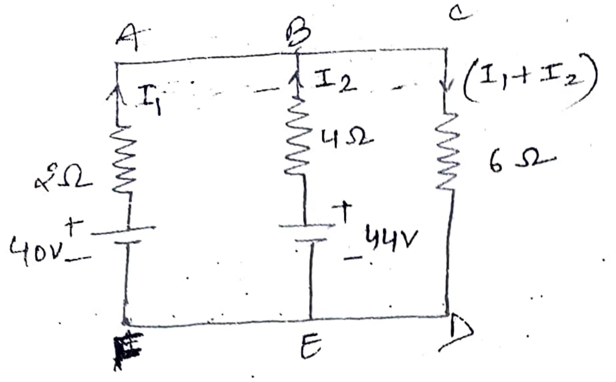



Fig. shows two batteries connected in parallel each represented by an emf along their internal resistances. A load resistance of 6Ω is connected across the ends of the batteries. Calculate current through each battery and load.




At node B Apply KCL

$\boxed{I_1 + I_2 = I_1 + I_2}$

$\cancel{\frac{40}{2}} + \cancel{\frac{44}{4}} \neq \cancel{I_1} + \cancel{I_2}$

$\cancel{20} + \cancel{11}$

Apply KVL in mesh ABEFA

$-40 + 2I_1 - 4I_2 + 44 = 0$

$2I_1 - 4I_2 + 4 = 0$

$I_1 - 2I_2 + 2 = 0$

$2I_2 - I_1 = 2$ — (1)

Apply KVL in BCDEB

$-44 + 4I_2 + 6(I_1 + I_2) = 0$

$-44 + 4I_2 + 6I_1 + 6I_2 = 0$

$6I_1 + 10I_2 = 44$

$3I_1 + 5I_2 = 22$ — (2)

from eqn (1)

$I_1 = 2I_2 - 2$ — (3)

Put $I_1$ in eqn (2)

$3(2I_2 - 2) + 5I_2 = 22$

$6I_2 - 6 + 5I_2 = 22$

$11I_2 = 28$

$\boxed{I_2 = \frac{28}{11} A}$

Put $I_2$ in eqn (1)

$2\left(\frac{28}{11}\right) - I_1 = 2$

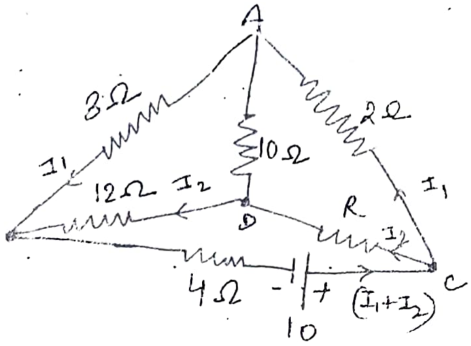
$\frac{56}{11} - 2 = I_1$

$I_1 = \frac{56 - 22}{11} = \frac{34}{11}$

$\boxed{I_1 = \frac{34}{11} A}$

Current throug load $= I_1 + I_2 = 62$

6/ Find the value of resistance R and current the branch AD carries no current.




$R = ?$

$I_R = I_2 = ?$

In mesh ABDA

$$3I_1 - 12I_2 = 0 \quad —(1)$$

In mesh ADCA

$$2I_1 - RI_2 = 0 \quad —(2)$$

from eqn (1)

$$3I_1 = 12I_2$$

$$I_1 = 4I_2 \quad —(3)$$

Put $I_1$ in eqn (2)

$$2(4I_2) - RI_2 = 0$$

$$8I_2 - RI_2 = 0$$

$$(8-R)I_2 = 0$$

$$8 - R = 0$$

$$\boxed{R = 8\Omega}$$

Apply KVL in mesh

$$-10 + 4(I_1+I_2) - \ldots$$

$$10 = 4I_1 + 4I_2 - \ldots$$

$$10 = 4I_1 + 16I_2 + 8I_2$$

$$10 = 4I_1 + 24I_2$$

$$5 = 2I_1 + 12I_2 \quad —(4)$$

Put eqn (3) in (4)

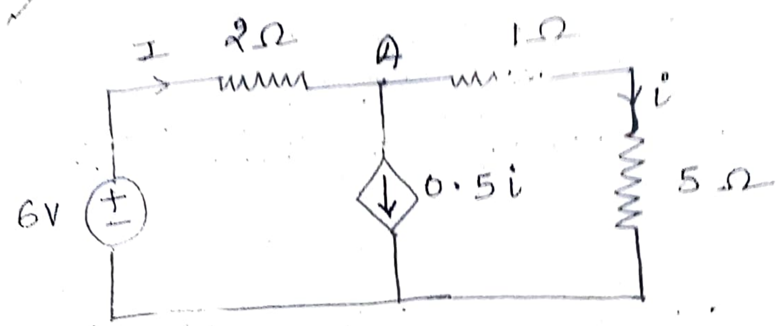
$$5 = 2(4I_2) + 12I_2$$

$$5 = 8I_2 + 12I_2$$

$$5 = 20I_2$$

$$\boxed{I_2 = \frac{5}{20} = 0.25\ A}$$

Q) Determine the current through 5 Ω resistor.




Apply KCL at node A.

$$I = 0.5i + i$$

$$I = 1.5i \quad \text{— (1)}$$

Apply KVL in outer loop,

$$-6 + 2I + i + 5i = 0$$

$$-6 + 2I + 6i = 0 \quad \text{— (2)}$$

Put (1) in (2):

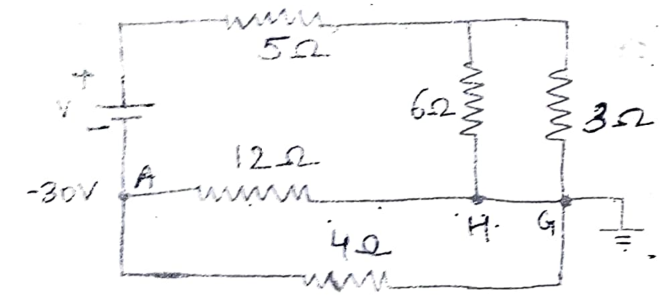
$$-6 + 2(1.5i) + 6i = 0$$

$$-6 + 3.0i + 6i = 0$$

$$6 = 9i$$

$$\boxed{i = \frac{2}{3} A}$$

Q5. In fig. the potential of point A = −30V. Using K law find (a) value of V &
(b) power dissipated by 5Ω resistance.



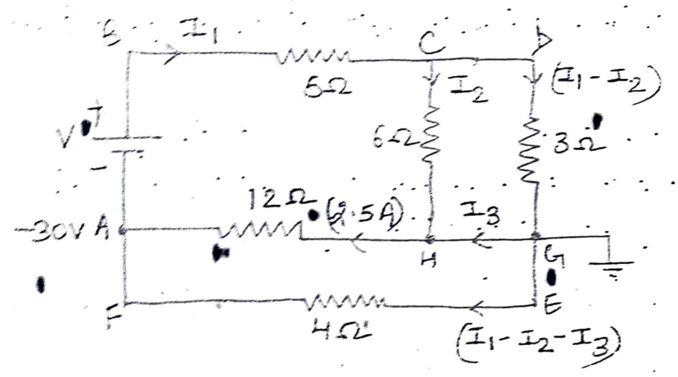

Soln potential at pt A = −30V
& at pt. G = 0V.

p.d. across 12Ω resistor connected between H & A
= 30V

current flow through 12Ω resistor = $\frac{30}{12}$ = 2.5A

& dirn is A ← H b/c current flows from higher potential to lower potential.

Before applying KCL & KVL, first indicate current in each branch.




At node H, Apply KCL $\quad I_2 + I_3 = 2.5$ — (1)

Apply KVL in ABCHA

$-V + 5I_1 + 6I_2 + 12(2.5) = 0$

$V = 5I_1 + 6I_2 + 30$ — (2)

Apply KVL in CDGH

$$3(I_1 - I_2) - 6I_2 = 0$$
$$3I_1 - 3I_2 - 6I_2 = 0$$
$$3I_1 - 9I_2 = 0$$
$$I_1 - 3I_2 = 0 \quad \text{--- } 3$$

Apply KVL in AHGEFA

$$4(I_1 - I_2 - I_3) - 12(2.5) = 0$$
$$4I_1 - 4I_2 - 4I_3 = 30 \quad \text{--- } 4$$

from 1

$$I_3 = 2.5 - I_2$$

Put $I_3$ in eq^n 4

$$4I_1 - 4I_2 - 4(2.5 - I_2) = 30$$
$$4I_1 - \cancel{4I_2} - 10.0 + \cancel{4I_2} = 30$$
$$4I_1 = 30 + 10$$
$$I_1 = \frac{40}{4} = 10A$$
$$\boxed{I_1 = 10A}$$

Put $I_1$ in eq^n 3

$$10 - 3I_2 = 0$$
$$\boxed{I_2 = \frac{10}{3}}$$

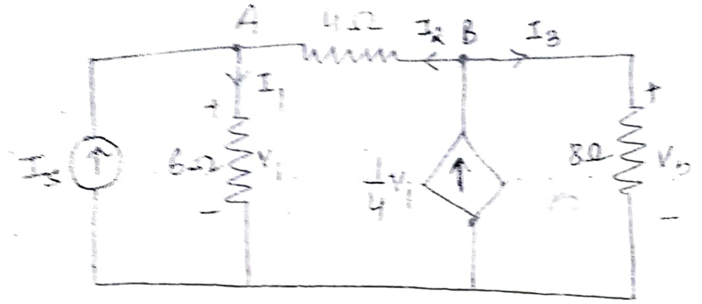
Put $I_1$ & $I_2$ in eq^n 2

$$V = 5(10) + 6\left(\frac{10}{3}\right) + 30$$
$$= 50 + 20 + 30$$

Power dissipated by $5\Omega$ resistance =

$$P = I_1^2 R$$
$$\boxed{P = 10 \times 10 \times 5 = 500W}$$

Applying Kirchhoff's current law, determine current $I_s$ in the electronic circuit. Take $V_o = 16V$




Soln At node A, - Apply KCL

$$I_s + I_2 = I_1$$

$$I_s + \frac{V_o - V_1}{4} = \frac{V_1}{6}$$

$$I_s + 4 - \frac{V_1}{4} = \frac{V_1}{6}$$

$$I_s + 4 = V_1\left(\frac{1}{6} + \frac{1}{4}\right)$$

$$I_s + 4 = \frac{5}{12}V_1 \quad \text{--- (1)}$$

At node B

$$\frac{1}{4}V_1 = I_2 + I_3$$

$$\frac{1}{4}V_1 = \frac{V_o - V_1}{4} + \frac{V_o}{8}$$

$$\frac{V_1}{4} = 4 - \frac{V_1}{4} + 2$$

$$\frac{V_1}{4} + \frac{V_1}{4} = 6$$

$$\frac{2V_1}{4} = 6$$

$$V_1 = 6 \times \frac{4}{2}$$

$$\boxed{V_1 = 12V}$$

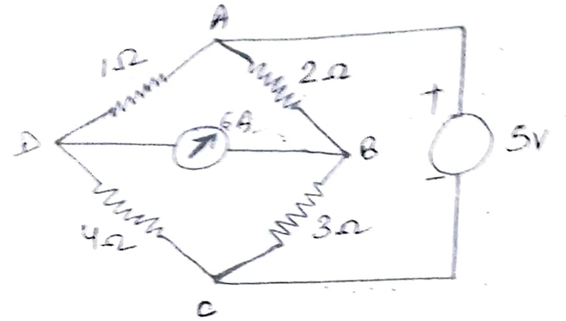
Put $V_1$ in eq (1)

$$I_s + 4 = 5\frac{12}{12}$$

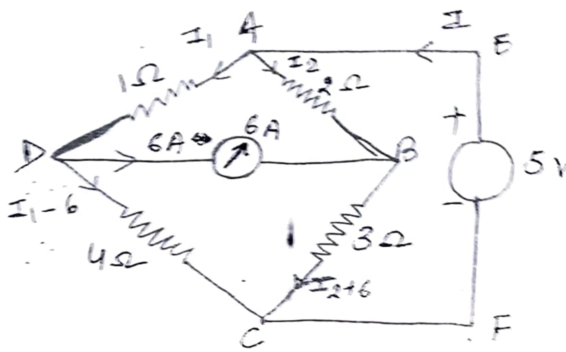
$$I_s = 5 - 4 = 1$$

$$\boxed{I_s = 1A}$$

Q. Find the total power delivered to the circuit by the sources.




Sol<sup></sup>




Apply KVL in AEFCBA,

$$-5 + 2I_2 + 3(I_2+6) = 0$$

$$2I_2 + 3I_2 + 18 - 5 = 0$$

$$5I_2 + 13 = 0$$

$$I_2 = -\frac{13}{5} A = -2.6 A \quad \text{--- (1)}$$

Apply KVL in ABCDA,

$$2I_2 + 3(I_2+6) - 4(I_1-6) - I_1 = 0$$

$$2I_2 + 3I_2 + 18 - 4I_1 + 24 - I_1 = 0$$

$$5I_2 - 5I_1 + 42 = 0$$

$$5\left[\frac{-13}{5}\right] - 5I_1 = -42$$

$$-13 + 42 = 5I_1$$

$$\frac{29}{5} = I_1 = 5.8 A \quad \text{--- (2)}$$

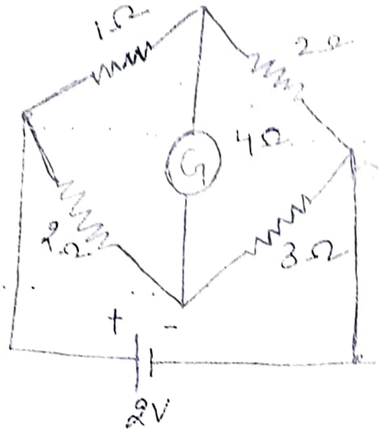
Total power delivered by two sources = total power absorbed by the circuit =

$$(I_2^2 \times 2) + (I_2+6)^2 \times 3 + (I_1-6)^2 \times 4 + I_1^2 \times 1$$

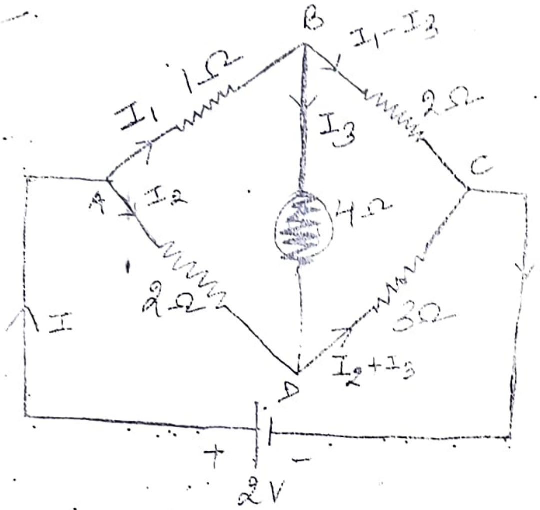
$$= (2.6)^2 \times 2 + (-2.6+6)^2 \times 3 + (5.8-6)^2 \times 4 + (5.8 ...$$

$$= \underline{82 W}$$

Calculate the current through the galvanometer in following bridge.




Soln




Apply KVL in mesh ABDA

$$I_1 + 4I_3 - 2I_2 = 0 \quad (1)$$

Apply KVL in mesh BCDB

$$2(I_1 - I_3) - 3(I_2 + I_3) - 4I_3 = 0$$

$$2I_1 - 2I_3 - 3I_2 - 3I_3 - 4I_3 = 0$$

$$2I_1 - 3I_2 - 9I_3 = 0 \quad (2)$$

In mesh ADCA

$$-2 + 2I_2 + 3(I_2 + I_3) = 0$$

$$-2 + 2I_2 + 3I_2 + 3I_3 = 0$$

$$-2 + 5I_2 + 3I_3 = 0$$

from ③

$-2+5I_2+3I_3=0$

$3I_3 = 2-5I_2$

$I_3 = \left(\frac{2-5I_2}{3}\right)$

Put $I_3$ in ①

$I_1+4I_3-2I_2=0$

$I_1+4\left(\frac{2-5I_2}{3}\right)-2I_2=0$

$I_1+\frac{(8-20I_2)}{3}-2I_2=0$

$3I_1+8-20I_2-6I_2=0$

$3I_1+8-26I_2=0$

$3I_1-26I_2=-8$ — ④

$3I_1 = 26I_2-8$

$I_1 = \frac{26I_2-8}{3}$ → ⑤

Put $I_1$ & $I_3$ in eqn ②

$2\left(\frac{26I_2-8}{3}\right)-3I_2-9\left(\frac{2-5I_2}{3}\right)=0$

$\frac{52I_2-16}{3}-3I_2-\frac{(18-45I_2)}{3}=0$

$52I_2-16-9I_2-18+45I_2=0$

$43I_2+45I_2-34=0$

$88I_2=34$

$I_2=+\frac{34}{88}A = \frac{17}{44}A$

Put $I_2$ in ⑤

$I_1 = \frac{26\left(\frac{17}{44}\right)-8}{3}$

$= \frac{13\cdot17-8\times22}{22\times3}$

$= \frac{221-176}{66}$

$=\frac{45}{66}=\frac{15}{22}=\frac{15}{22}A$

$=\frac{\left(26\times\frac{17}{44}\right)-8}{3}$

$=\frac{26\times17-8\times44}{3\times44}$

$=\frac{442-}{3\times4}$

Put $I_2$ in eqn of $I_3$

$I_3 = \frac{2-5\left(\frac{17}{44}\right)}{3}$

$=\frac{88-85}{44\times3}$

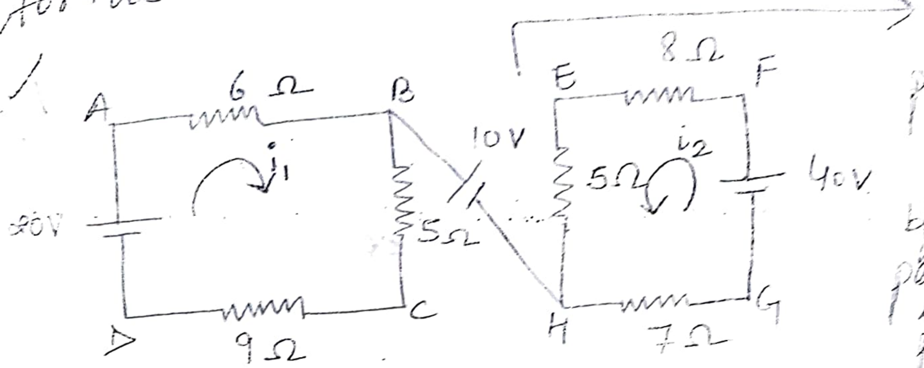
$=\frac{3}{44\times3}=\frac{1}{44}$

$\boxed{I_3=\frac{1}{44}}$

13 × 17, 91, 13, 221, 176, 45

Numericals from BLT

Q.11 For the circuit shown in fig. find $V_{CE}$ and $V_{AG}$




No path ... by 10V ... KC for closed path → current leave from point and enter the same point. 10V ... a potential difference ... two points BA

Let $i_1$ be the current flow in mesh ABCD.

$$i_1 = \frac{20}{6+5+9} = \frac{20}{20} = 1$$

$$\boxed{i_1 = 1A}$$

Let '$i_2$' be the current flow in mesh EFGH

$$i_2 = \frac{40}{5+7+8} = \frac{40}{20} = 2$$

$$\boxed{i_2 = 2A}$$

for $V_{CE}$, we can find out the value by taking the algebraic sum of the voltage drops met on the way from point ~~E to~~ C to E.

~~$V_{CE} = (2\times5) - 10 + (5\times1) = 10+10+5$~~

$$V_{CE} = (-5\times1) + 10 - (5\times2) = -5 + \cancel{10} - \cancel{10}$$

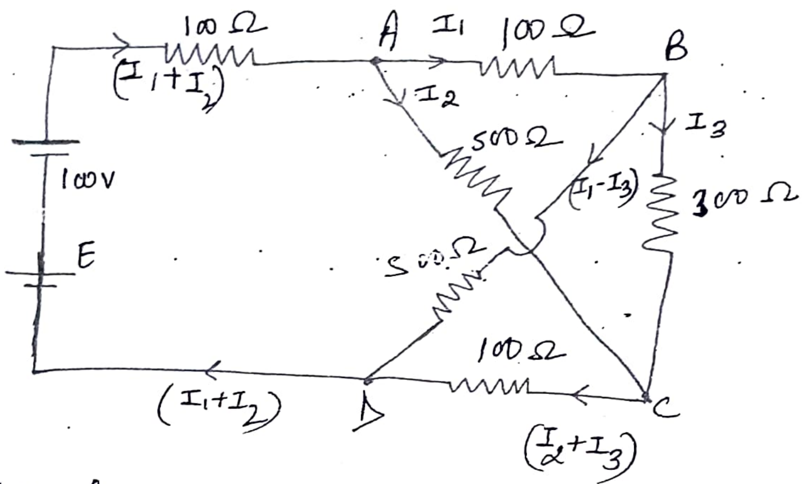
$$\boxed{V_{CE} = -5V.}$$

Ily, for $V_{AG}$, we proceed from A to G

$$V_{AG} = (6\times1) + 10 + (7\times2) = 6+10+14$$
$$= 30V$$

$$\boxed{V_{AG} = 30V}$$

Q. 12 Determine the current supplied by the bat in the circuit as shown in fig.




Apply KVL in mesh ABCA,

$\cancel{100 I_1 + 100 I_3 + 100(I_2+I_3) - 500(I_1-I_3) = 0}$

$100 I_1 + 300 I_3 - 500 I_2 = 0$

$100(I_1 + 3I_3 - 5I_2) = 0$

$I_1 + 3I_3 - 5I_2 = 0$ —— (1)

Apply KVL in mesh BCDB

$300 I_3 + 100(I_2+I_3) - 500(I_1 - I_3) = 0$

$300 I_3 + 100 I_2 + 100 I_3 - 500 I_1 + 500 I_3 = 0$

$-500 I_1 + 100 I_2 + 900 I_3 = 0$

$-5I_1 + I_2 + 9I_3 = 0$ —— (2)

$5I_1 - I_2 - 9I_3 = 0$ —— (2)

Apply KVL in mesh ABDEA

$100 I_1 + 500(I_1 - I_3) - 100 + 100(I_1 + I_2) = 0$

$100 I_1 + 500 I_1 - 500 I_3 - 100 + 100 I_1 + 100 I_2 = 0$

$700 I_1 + 100 I_2 - 500 I_3 = 100$

$100(7I_1 + I_2 - 5I_3) = 100$

Consider 3 eqⁿs

$I_1 - 5I_2 + 3I_3 = 0$ —— (1)
$5I_1 - I_2 - 9I_3 = 0$ —— (2)
$7I_1 + I_2 - 5I_3 = 1$ —— (3)

Method 1 Solve by using Matrix Method (Cramer's Ru...

$$\begin{bmatrix} 1 & -5 & 3 \\ 5 & -1 & -9 \\ 7 & 1 & -5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 1 & -5 & 3 \\ 5 & -1 & -9 \\ 7 & 1 & -5 \end{bmatrix} = 1(5-(-9)) + 5(-25+63) + 3(5+7)$$

$$= 14 + 5(38) + 3(12)$$

$$= 14 + 190 + 36$$

$$\boxed{\Delta = 240}$$

$$\Delta_1 = \begin{bmatrix} 0 & -5 & 3 \\ 0 & -1 & -9 \\ 1 & 1 & -5 \end{bmatrix} = 0 + 5(0+9) + 3(0+1)$$

$$= 45 + 3$$

$$\boxed{\Delta_1 = 48}$$

$$\Delta_2 = \begin{bmatrix} 1 & 0 & 3 \\ 5 & 0 & -9 \\ 7 & 1 & -5 \end{bmatrix} = 1(0+9) - 0 + 3(5)$$

$$= 9 + 15$$

$$\boxed{\Delta_2 = 24}$$

$$\Delta_3 = \begin{bmatrix} 1 & -5 & 0 \\ 5 & -1 & 0 \\ 7 & 1 & 1 \end{bmatrix}$$

$$= 1(-1) + 5(5-0) + 0$$

$$= -1 + 25$$

$$\boxed{\Delta_3 = 24}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{\cancel{48}\ \cancel{24}\ \cancel{12}\ \cancel{6}\ 3}{\cancel{240}\ \cancel{120}\ \cancel{60}\ \cancel{30}\ 15} = \frac{1}{5} A$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{24}{240} = \frac{1}{10} A$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{24}{240} = \frac{1}{10} A$$

$$I = (I_1 + I_2) = \frac{1}{5} + \frac{1}{10} = \frac{2+1}{10} = \frac{3}{10} A$$

<u>Method-2:</u>

Consider eq$^n$ (3)

$$7I_1 + I_2 - 5I_3 = 1$$

<u>Obtain $I_3$</u>

$$7I_1 + I_2 - 1 = 5I_3$$

$$I_3 = \left(\frac{7I_1 + I_2 - 1}{5}\right)$$